7.1 Rules That Fail for Matrices

WORKSHEET

The question

STEP II 1993, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself For real numbers, why is it true that $ab = 0$ implies $a = 0$ or $b = 0$? Which property of numbers does the argument use, and does it hold for matrices?

STAGE 1 Can $\mathbf{AB} = \mathbf{0}$ but $\mathbf{BA} \neq \mathbf{0}$?

Ask yourself Think geometrically. You want $\mathbf{B}$ to send everything into a line that $\mathbf{A}$ then squashes to zero. But in the other order, $\mathbf{A}$ should not squash everything that $\mathbf{B}$ then kills.

STAGE 3 Non-zero solutions of $\mathbf{AX} = \mathbf{0}$

Ask yourself If $\det \mathbf{A} \neq 0$, what can you do to $\mathbf{AX} = \mathbf{0}$? If $\det \mathbf{A} = 0$, can you find a non-zero column vector $\mathbf{v}$ with $\mathbf{Av} = \mathbf{0}$?

STAGE 4 How many square roots?

Pause and try Find non-zero $\mathbf{A}$ and $\mathbf{B}$ for which $\mathbf{X}^2 + \mathbf{AX} + \mathbf{B} = \mathbf{0}$ has infinitely many solutions. (*Hint:* make the left-hand side a perfect square, $(\mathbf{X} - \mathbf{I})^2$, and think about matrices whose square is zero.)

Try this next

1. **When do they commute?** Find all 2×2 matrices that commute with $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

2. **Square roots of zero.** Find all 2×2 real matrices $\mathbf{N}$ with $\mathbf{N}^2 = \mathbf{0}$. (*Hint:* use the Cayley–Hamilton theorem to show $\operatorname{tr} \mathbf{N} = \det \mathbf{N} = 0$.)

3. **Reflections.** Show that every reflection matrix $\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ squares to $\mathbf{I}$, and find the line it reflects in.

4. **Cayley–Hamilton.** Prove that every 2×2 matrix satisfies $\mathbf{M}^2 - (\operatorname{tr} \mathbf{M})\mathbf{M} + (\det \mathbf{M})\mathbf{I} = \mathbf{0}$ by direct calculation.

Further reading

- **Gilbert Strang,** ***Introduction to Linear Algebra*** (Wellesley-Cambridge). The most popular first linear algebra text, with an emphasis on understanding.
- **Sheldon Axler,** ***Linear Algebra Done Right*** (Springer; free online). A more abstract approach, centred on linear maps rather than matrices.
- **Jim Baggott,** ***The Quantum Story*** (Oxford University Press). The history of quantum mechanics, including Heisenberg's matrix mechanics.

The full guided solution, and what the question leads to at university, is in Section 7.1:
https://beyondstep.keithmcnulty.org/ch07/matrix-rules.html